- *Elliptic curve cryptography (ECC)* **--**At the protocol level, blockchain protocol uses the ECC family of algorithms instead of the traditional RSA (Rivest-Shamir-Edelman) algorithm. Why ECC, not RSA? ECC is stronger than RSA for a given number of bits. A 256-bit ECC key-pair is equivalent in strength to about a 3072-bit RSA key pair.
- *Tx and block hashes* --Tx hashes and block hashes are computed at the time of their creation. Any modifications (even a single bit) to a Tx or a block will result in a mismatched hash, resulting in rejection of the Tx and the block securing the integrity of the chain.
- *Off-chain data security* --Application data off-chain can be secured by hashing the data and storing *only* the hash value on the chain. This concept is discussed in chapters 2-6, in which the airline data off-chain can be secured by a hash on the chain.
- *On-chain data security* --At the application level, a combination of encryption, hashing, and one-time-password helps secure the data transmitted in a Tx. You used hashing and encryption with a one-time password to secure the bid in the MPC-Dapp in chapters 7 and 8.

Thus, a combination of hashing and cryptography plays a critical role in the blockchain creation process, as well as the integrity of transactions and the security of data. A developer will routinely use cryptography and hashing algorithms in their development. As you learned in chapter 9, the web3 API provides Keccak and SHA3 functions to facilitate security in Dapps.

Do you wonder about the role of cryptocurrency in a typical Dapp? Do you wonder whether you can develop without the use